

2520090076

My SQL Bank views

A **View** in SQL is a virtual table created from the result of a SQL query. It does not store data physically; instead, it displays data from one or more underlying tables. Views help simplify complex queries, improve security by restricting access to specific columns or rows, and provide a customized representation of data for users. Views can be created using simple SELECT statements, JOINS, aggregate functions, GROUP BY, HAVING, and other SQL clauses. Some views are updatable, allowing INSERT, UPDATE, and DELETE operations on the underlying tables. Views are widely used in database applications to improve data organization, maintainability, and ease of access.

```
mysql> USE BankViewsDB;
Database changed
mysql> CREATE TABLE Customer (
->     Customer_ID INT PRIMARY KEY,
->     Customer_Name VARCHAR(100) NOT NULL,
->     Phone VARCHAR(15),
->     Email VARCHAR(100),
->     City VARCHAR(50)
-> );
Query OK, 0 rows affected (0.12 sec)

mysql> CREATE TABLE Account (
->     Account_No INT PRIMARY KEY,
->     Customer_ID INT,
->     Account_Type VARCHAR(20),
->     Balance DECIMAL(12,2),
->     Branch VARCHAR(50),
->
->     FOREIGN KEY (Customer_ID)
->     REFERENCES Customer(Customer_ID)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE Bank_Transaction (
->     Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
->     Account_No INT,
->     Transaction_Type VARCHAR(20),
->     Amount DECIMAL(12,2),
->     Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
->
->     FOREIGN KEY (Account_No)
->     REFERENCES Account(Account_No)
-> );
```

```
mysql> CREATE TABLE Loan (  
->     Loan_ID INT PRIMARY KEY,  
->     Customer_ID INT,  
->     Loan_Type VARCHAR(30),  
->     Loan_Amount DECIMAL(12,2),  
->     Interest_Rate DECIMAL(5,2),  
->  
->     FOREIGN KEY (Customer_ID)  
->     REFERENCES Customer(Customer_ID)  
-> );
```

Query OK, 0 rows affected (0.04 sec)

```
mysql> INSERT INTO Customer  
-> (Customer_ID, Customer_Name, Phone, Email, City)  
-> VALUES  
-> (101, 'Ravi Kumar', '9876543210', 'ravi@gmail.com', 'Hyderabad'),  
-> (102, 'Priya Sharma', '9876543211', 'priya@gmail.com', 'Vijayawada'),  
-> (103, 'Arjun Reddy', '9876543212', 'arjun@gmail.com', 'Bangalore'),  
-> (104, 'Sneha Rao', '9876543213', 'sneha@gmail.com', 'Chennai'),  
-> (105, 'Kiran Kumar', '9876543214', 'kiran@gmail.com', 'Hyderabad'),  
-> (106, 'Anil Kumar', '9876543215', 'anil@gmail.com', 'Delhi'),  
-> (107, 'Meena Reddy', '9876543216', 'meena@gmail.com', 'Mumbai'),  
-> (108, 'Rahul Sharma', '9876543217', 'rahul@gmail.com', 'Pune'),  
-> (109, 'Lakshmi Devi', '9876543218', 'lakshmi@gmail.com', 'Hyderabad'),  
-> (110, 'Suresh Babu', '9876543219', 'suresh@gmail.com', 'Vijayawada');
```

Query OK, 10 rows affected (0.02 sec)

Records: 10 Duplicates: 0 Warnings: 0

```
mysql> INSERT INTO Account
-> (Account_No, Customer_ID, Account_Type, Balance, Branch)
-> VALUES
-> (10001, 101, 'Savings', 50000, 'Hyderabad'),
-> (10002, 102, 'Savings', 75000, 'Vijayawada'),
-> (10003, 103, 'Current', 120000, 'Bangalore'),
-> (10004, 104, 'Savings', 45000, 'Chennai'),
-> (10005, 105, 'Current', 90000, 'Hyderabad'),
-> (10006, 106, 'Savings', 65000, 'Delhi'),
-> (10007, 107, 'Current', 150000, 'Mumbai'),
-> (10008, 108, 'Savings', 35000, 'Pune'),
-> (10009, 109, 'Savings', 85000, 'Hyderabad'),
-> (10010, 110, 'Current', 110000, 'Vijayawada');
```

Query OK, 10 rows affected (0.01 sec)
Records: 10 Duplicates: 0 Warnings: 0

```
mysql> INSERT INTO Bank_Transaction
-> (Account_No, Transaction_Type, Amount)
-> VALUES
-> (10001, 'DEPOSIT', 10000),
-> (10001, 'WITHDRAW', 5000),
-> (10002, 'DEPOSIT', 15000),
-> (10003, 'WITHDRAW', 20000),
-> (10004, 'DEPOSIT', 5000),
-> (10005, 'WITHDRAW', 10000),
-> (10006, 'DEPOSIT', 12000),
-> (10007, 'DEPOSIT', 25000),
-> (10008, 'WITHDRAW', 5000),
-> (10009, 'DEPOSIT', 20000),
-> (10010, 'WITHDRAW', 15000);
```

Query OK, 11 rows affected (0.01 sec)
Records: 11 Duplicates: 0 Warnings: 0

```
mysql> INSERT INTO Loan
-> (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate)
-> VALUES
-> (501, 101, 'Home Loan', 5000000, 7.5),
-> (502, 102, 'Education Loan', 1000000, 6.5),
-> (503, 103, 'Car Loan', 800000, 8.2),
-> (504, 104, 'Personal Loan', 500000, 10.5),
-> (505, 105, 'Home Loan', 4000000, 7.2),
-> (506, 106, 'Car Loan', 900000, 8.5),
-> (507, 107, 'Business Loan', 3000000, 9.0),
-> (508, 109, 'Personal Loan', 600000, 10.0);
```

Query OK, 8 rows affected (0.02 sec)
Records: 8 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM Customer;
```

Customer_ID	Customer_Name	Phone	Email	City
101	Ravi Kumar	9876543210	ravi@gmail.com	Hyderabad
102	Priya Sharma	9876543211	priya@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad
106	Anil Kumar	9876543215	anil@gmail.com	Delhi
107	Meena Reddy	9876543216	meena@gmail.com	Mumbai
108	Rahul Sharma	9876543217	rahul@gmail.com	Pune
109	Lakshmi Devi	9876543218	lakshmi@gmail.com	Hyderabad
110	Suresh Babu	9876543219	suresh@gmail.com	Vijayawada

10 rows in set (0.01 sec)

```
mysql> SELECT * FROM Account;
```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
10006	106	Savings	65000.00	Delhi
10007	107	Current	150000.00	Mumbai
10008	108	Savings	35000.00	Pune
10009	109	Savings	85000.00	Hyderabad
10010	110	Current	110000.00	Vijayawada

10 rows in set (0.00 sec)

```
mysql> SELECT * FROM Bank_Transaction;
```

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-15 16:02:47
2	10001	WITHDRAW	5000.00	2026-08-15 16:02:47
3	10002	DEPOSIT	15000.00	2026-08-15 16:02:47
4	10003	WITHDRAW	20000.00	2026-08-15 16:02:47
5	10004	DEPOSIT	5000.00	2026-08-15 16:02:47
6	10005	WITHDRAW	10000.00	2026-08-15 16:02:47
7	10006	DEPOSIT	12000.00	2026-08-15 16:02:47
8	10007	DEPOSIT	25000.00	2026-08-15 16:02:47
9	10008	WITHDRAW	5000.00	2026-08-15 16:02:47
10	10009	DEPOSIT	20000.00	2026-08-15 16:02:47
11	10010	WITHDRAW	15000.00	2026-08-15 16:02:47

```
mysql> SELECT * FROM Loan;
```

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
501	101	Home Loan	5000000.00	7.50
502	102	Education Loan	1000000.00	6.50
503	103	Car Loan	800000.00	8.20
504	104	Personal Loan	500000.00	10.50
505	105	Home Loan	4000000.00	7.20
506	106	Car Loan	900000.00	8.50
507	107	Business Loan	3000000.00	9.00
508	109	Personal Loan	600000.00	10.00

```
8 rows in set (0.00 sec)
```

```
mysql> CREATE VIEW Customer_View AS
```

```
-> SELECT *
```

```
-> FROM Customer;
```

```
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT * FROM Customer_View;
```

Customer_ID	Customer_Name	Phone	Email	City
101	Ravi Kumar	9876543210	ravi@gmail.com	Hyderabad
102	Priya Sharma	9876543211	priya@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad
106	Anil Kumar	9876543215	anil@gmail.com	Delhi
107	Meena Reddy	9876543216	meena@gmail.com	Mumbai
108	Rahul Sharma	9876543217	rahul@gmail.com	Pune
109	Lakshmi Devi	9876543218	lakshmi@gmail.com	Hyderabad
110	Suresh Babu	9876543219	suresh@gmail.com	Vijayawada

```
10 rows in set (0.01 sec)
```

```
mysql> CREATE VIEW Customer_Basic_View AS
-> SELECT
->     Customer_ID,
->     Customer_Name,
->     City
-> FROM Customer;
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> SELECT * FROM Customer_Basic_View;
```

Customer_ID	Customer_Name	City
101	Ravi Kumar	Hyderabad
102	Priya Sharma	Vijayawada
103	Arjun Reddy	Bangalore
104	Sneha Rao	Chennai
105	Kiran Kumar	Hyderabad
106	Anil Kumar	Delhi
107	Meena Reddy	Mumbai
108	Rahul Sharma	Pune
109	Lakshmi Devi	Hyderabad
110	Suresh Babu	Vijayawada

10 rows in set (0.00 sec)

```
mysql> CREATE VIEW Account_View AS
-> SELECT
->     Account_No,
->     Account_Type,
->     Balance,
->     Branch
-> FROM Account;
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> SELECT * FROM Account_View;
```

Account_No	Account_Type	Balance	Branch
10001	Savings	50000.00	Hyderabad
10002	Savings	75000.00	Vijayawada
10003	Current	120000.00	Bangalore
10004	Savings	45000.00	Chennai
10005	Current	90000.00	Hyderabad
10006	Savings	65000.00	Delhi
10007	Current	150000.00	Mumbai
10008	Savings	35000.00	Pune
10009	Savings	85000.00	Hyderabad
10010	Current	110000.00	Vijayawada

10 rows in set (0.00 sec)

```
mysql> CREATE VIEW Savings_Account_View AS
-> SELECT *
-> FROM Account
-> WHERE Account_Type = 'Savings';
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> SELECT * FROM Savings_Account_View;
```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10004	104	Savings	45000.00	Chennai
10006	106	Savings	65000.00	Delhi
10008	108	Savings	35000.00	Pune
10009	109	Savings	85000.00	Hyderabad

6 rows in set (0.00 sec)

```
mysql> CREATE VIEW Current_Account_View AS
-> SELECT *
-> FROM Account
-> WHERE Account_Type = 'Current';
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> SELECT * FROM Current_Account_View;
+-----+-----+-----+-----+-----+
| Account_No | Customer_ID | Account_Type | Balance | Branch |
+-----+-----+-----+-----+-----+
| 10003 | 103 | Current | 120000.00 | Bangalore |
| 10005 | 105 | Current | 90000.00 | Hyderabad |
| 10007 | 107 | Current | 150000.00 | Mumbai |
| 10010 | 110 | Current | 110000.00 | Vijayawada |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Current_Account_View;
+-----+-----+-----+-----+-----+
| Account_No | Customer_ID | Account_Type | Balance | Branch |
+-----+-----+-----+-----+-----+
| 10003 | 103 | Current | 120000.00 | Bangalore |
| 10005 | 105 | Current | 90000.00 | Hyderabad |
| 10007 | 107 | Current | 150000.00 | Mumbai |
| 10010 | 110 | Current | 110000.00 | Vijayawada |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> CREATE VIEW High_Balance_View AS
-> SELECT
-> Account_No,
-> Customer_ID,
-> Account_Type,
-> Balance
-> FROM Account
```

```
mysql> SELECT * FROM High_Balance_View;
+-----+-----+-----+-----+
| Account_No | Customer_ID | Account_Type | Balance |
+-----+-----+-----+-----+
| 10003 | 103 | Current | 120000.00 |
| 10007 | 107 | Current | 150000.00 |
| 10010 | 110 | Current | 110000.00 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> CREATE VIEW Hyderabad_Account_View AS
-> SELECT *
-> FROM Account
-> WHERE Branch = 'Hyderabad';
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT * FROM Hyderabad_Account_View;
+-----+-----+-----+-----+-----+
| Account_No | Customer_ID | Account_Type | Balance | Branch |
+-----+-----+-----+-----+-----+
| 10001 | 101 | Savings | 50000.00 | Hyderabad |
| 10005 | 105 | Current | 90000.00 | Hyderabad |
| 10009 | 109 | Savings | 85000.00 | Hyderabad |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> CREATE VIEW Low_Balance_View AS
-> SELECT
-> Account_No,
-> Customer_ID,
-> Balance
-> FROM Account
-> WHERE Balance < 50000;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT * FROM Low_Balance_View;
```

```
mysql> SELECT * FROM Low_Balance_View;
```

Account_No	Customer_ID	Balance
10004	104	45000.00
10008	108	35000.00

2 rows in set (0.00 sec)

```
mysql> CREATE VIEW Customer_Account_View AS
```

```
-> SELECT
->   C.Customer_ID,
->   C.Customer_Name,
->   C.City,
->   A.Account_No,
->   A.Account_Type,
->   A.Balance,
->   A.Branch
-> FROM Customer C
-> JOIN Account A
-> ON C.Customer_ID = A.Customer_ID;
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> SELECT * FROM Customer_Account_View;
```

Customer_ID	Customer_Name	City	Account_No	Account_Type	Balance	Branch
101	Ravi Kumar	Hyderabad	10001	Savings	50000.00	Hyderabad
102	Priya Sharma	Vijayawada	10002	Savings	75000.00	Vijayawada
103	Arjun Reddy	Bangalore	10003	Current	120000.00	Bangalore
104	Sneha Rao	Chennai	10004	Savings	45000.00	Chennai
105	Kiran Kumar	Hyderabad	10005	Current	90000.00	Hyderabad
106	Anil Kumar	Delhi	10006	Savings	65000.00	Delhi
107	Meena Reddy	Mumbai	10007	Current	150000.00	Mumbai
108	Rahul Sharma	Pune	10008	Savings	35000.00	Pune
109	Lakshmi Devi	Hyderabad	10009	Savings	85000.00	Hyderabad
110	Suresh Babu	Vijayawada	10010	Current	110000.00	Vijayawada

```
mysql> SELECT * FROM Customer_Account_View;
```

Customer_ID	Customer_Name	City	Account_No	Account_Type	Balance	Branch
101	Ravi Kumar	Hyderabad	10001	Savings	50000.00	Hyderabad
102	Priya Sharma	Vijayawada	10002	Savings	75000.00	Vijayawada
103	Arjun Reddy	Bangalore	10003	Current	120000.00	Bangalore
104	Sneha Rao	Chennai	10004	Savings	45000.00	Chennai
105	Kiran Kumar	Hyderabad	10005	Current	90000.00	Hyderabad
106	Anil Kumar	Delhi	10006	Savings	65000.00	Delhi
107	Meena Reddy	Mumbai	10007	Current	150000.00	Mumbai
108	Rahul Sharma	Pune	10008	Savings	35000.00	Pune
109	Lakshmi Devi	Hyderabad	10009	Savings	85000.00	Hyderabad
110	Suresh Babu	Vijayawada	10010	Current	110000.00	Vijayawada

10 rows in set (0.02 sec)

```
mysql> CREATE VIEW Hyderabad_Customer_Accounts AS
```

```
-> SELECT
->   C.Customer_Name,
->   C.City,
->   A.Account_No,
->   A.Account_Type,
->   A.Balance
-> FROM Customer C
-> JOIN Account A
-> ON C.Customer_ID = A.Customer_ID
-> WHERE A.Branch = 'Hyderabad';
```

Query OK, 0 rows affected (0.01 sec)


```
mysql> SELECT * FROM Hyderabad_Customer_Accounts;
```

Customer_Name	City	Account_No	Account_Type	Balance
Ravi Kumar	Hyderabad	10001	Savings	50000.00
Kiran Kumar	Hyderabad	10005	Current	90000.00
Lakshmi Devi	Hyderabad	10009	Savings	85000.00

3 rows in set (0.00 sec)

```
mysql> CREATE VIEW Customer_Loan_View AS
-> SELECT
-> C.Customer_ID,
-> C.Customer_Name,
-> C.City,
-> L.Loan_ID,
-> L.Loan_Type,
-> L.Loan_Amount,
-> L.Interest_Rate
-> FROM Customer C
-> JOIN Loan L
-> ON C.Customer_ID = L.Customer_ID;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT * FROM Customer_Loan_View;
```

Customer_ID	Customer_Name	City	Loan_ID	Loan_Type	Loan_Amount	Interest_Rate
101	Ravi Kumar	Hyderabad	501	Home Loan	500000.00	7.50
102	Priya Sharma	Vijayawada	502	Education Loan	100000.00	6.50
103	Arjun Reddy	Bangalore	503	Car Loan	80000.00	8.20
104	Sneha Rao	Chennai	504	Personal Loan	50000.00	10.50
105	Kiran Kumar	Hyderabad	505	Home Loan	400000.00	7.20
106	Anil Kumar	Delhi	506	Car Loan	90000.00	8.50
107	Meena Reddy	Mumbai	507	Business Loan	300000.00	9.00
109	Lakshmi Devi	Hyderabad	508	Personal Loan	60000.00	10.00

8 rows in set (0.00 sec)

```
mysql> CREATE VIEW Customer_Banking_View AS
-> SELECT
-> C.Customer_ID,
-> C.Customer_Name,
-> C.City,
-> A.Account_No,
-> A.Account_Type,
-> A.Balance,
-> A.Branch,
-> L.Loan_Type,
-> L.Loan_Amount
-> FROM Customer C
-> LEFT JOIN Account A
-> ON C.Customer_ID = A.Customer_ID
-> LEFT JOIN Loan L
-> ON C.Customer_ID = L.Customer_ID;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT * FROM Customer_Banking_View;
```

Customer_ID	Customer_Name	City	Account_No	Account_Type	Balance	Branch	Loan_Type	Loan_Amount
101	Ravi Kumar	Hyderabad	10001	Savings	50000.00	Hyderabad	Home Loan	500000.00
102	Priya Sharma	Vijayawada	10002	Savings	75000.00	Vijayawada	Education Loan	100000.00
103	Arjun Reddy	Bangalore	10003	Current	120000.00	Bangalore	Car Loan	80000.00
104	Sneha Rao	Chennai	10004	Savings	45000.00	Chennai	Personal Loan	50000.00
105	Kiran Kumar	Hyderabad	10005	Current	90000.00	Hyderabad	Home Loan	400000.00
106	Anil Kumar	Delhi	10006	Savings	65000.00	Delhi	Car Loan	90000.00
107	Meena Reddy	Mumbai	10007	Current	150000.00	Mumbai	Business Loan	300000.00
108	Rahul Sharma	Pune	10008	Savings	35000.00	Pune	NULL	NULL
109	Lakshmi Devi	Hyderabad	10009	Savings	85000.00	Hyderabad	Personal Loan	60000.00
110	Suresh Babu	Vijayawada	10010	Current	110000.00	Vijayawada	NULL	NULL

10 rows in set (0.00 sec)

```
mysql> CREATE VIEW Total_Bank_Balance AS
-> SELECT
-> SUM(Balance) AS Total_Balance
-> FROM Account;
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> SELECT * FROM Total_Bank_Balance;
```

Total_Balance
825000.00

1 row in set (0.03 sec)

```
mysql> CREATE VIEW Average_Account_Balance AS
-> SELECT
```

```
mysql> SELECT * FROM Average_Account_Balance;
+-----+
| Average_Balance |
+-----+
|      82500.000000 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> CREATE VIEW Maximum_Balance_View AS
-> SELECT
->     MAX(Balance) AS Maximum_Balance
-> FROM Account;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT * FROM Maximum_Balance_View;
+-----+
| Maximum_Balance |
+-----+
|      150000.00 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> SELECT * FROM Maximum_Balance_View;
+-----+
| Maximum_Balance |
+-----+
|      150000.00 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> CREATE VIEW Minimum_Balance_View AS
-> SELECT
->     MIN(Balance) AS Minimum_Balance
-> FROM Account;
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE VIEW Minimum_Balance_View AS
-> SELECT
->     MIN(Balance) AS Minimum_Balance
-> FROM Account;
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> SELECT * FROM Minimum_Balance_View;
+-----+
| Minimum_Balance |
+-----+
|       35000.00 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> CREATE VIEW Account_Count_View AS
-> SELECT
->     COUNT(*) AS Total_Accounts
-> FROM Account;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT * FROM Account_Count_View;
+-----+
| Total_Accounts |
+-----+
|           10 |
+-----+
1 row in set (0.02 sec)
```

```
mysql> CREATE VIEW Branch_Account_Count AS
-> SELECT
->     Branch,
->     COUNT(*) AS Number_of_Accounts
-> FROM Account
-> GROUP BY Branch;
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> SELECT * FROM Branch_Account_Count;
```

Branch	Number_of_Accounts
Hyderabad	3
Vijayawada	2
Bangalore	1
Chennai	1
Delhi	1
Mumbai	1
Pune	1

7 rows in set (0.02 sec)

```
mysql> CREATE VIEW Branch_Total_Balance AS
-> SELECT
->   Branch,
->   SUM(Balance) AS Total_Balance
-> FROM Account
-> GROUP BY Branch;
```

Query OK, 0 rows affected (0.02 sec)

```
mysql> SELECT * FROM Branch_Total_Balance;
```

Branch	Total_Balance
Hyderabad	225000.00
Vijayawada	185000.00
Bangalore	120000.00
Chennai	45000.00
Delhi	65000.00
Mumbai	150000.00
Pune	35000.00

7 rows in set (0.00 sec)

```
mysql> CREATE VIEW Account_Type_Count AS
-> SELECT
->   Account_Type,
->   COUNT(*) AS Number_of_Accounts
-> FROM Account
-> GROUP BY Account_Type;
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> SELECT * FROM Account_Type_Count;
```

Account_Type	Number_of_Accounts
Savings	6
Current	4

2 rows in set (0.00 sec)

```
mysql> CREATE VIEW Account_Type_Balance AS
-> SELECT
->   Account_Type,
->   SUM(Balance) AS Total_Balance
-> FROM Account
-> GROUP BY Account_Type;
```

Query OK, 0 rows affected (0.03 sec)

```
mysql> SELECT * FROM Account_Type_Balance;
```

Account_Type	Total_Balance
Savings	355000.00
Current	470000.00

2 rows in set (0.00 sec)

```
mysql> CREATE VIEW Multiple_Account_Branches AS
-> SELECT
```

```
mysql> CREATE VIEW Multiple_Account_Branches AS
-> SELECT
->     Branch,
->     COUNT(*) AS Number_of_Accounts
-> FROM Account
-> GROUP BY Branch
-> HAVING COUNT(*) > 1;
Query OK, 0 rows affected (0.02 sec)

mysql> SELECT * FROM Multiple_Account_Branches;
+-----+-----+
| Branch | Number_of_Accounts |
+-----+-----+
| Hyderabad | 3 |
| Vijayawada | 2 |
+-----+-----+
2 rows in set (0.00 sec)

mysql> CREATE VIEW Multiple_Account_Branches AS
-> SELECT
->     Branch,
->     COUNT(*) AS Number_of_Accounts
-> FROM Account
-> GROUP BY Branch
-> HAVING COUNT(*) > 1;
ERROR 1050 (42501): Table 'Multiple_Account_Branches' already exists
mysql> CREATE VIEW Rich_Branches AS
-> SELECT
->     Branch,
->     SUM(Balance) AS Total_Balance
-> FROM Account
-> GROUP BY Branch
-> HAVING SUM(Balance) > 100000;
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> SELECT * FROM Rich_Branches;
+-----+-----+
| Branch | Total_Balance |
+-----+-----+
```

```
mysql> SELECT * FROM Rich_Branches;
+-----+-----+
| Branch | Total_Balance |
+-----+-----+
| Hyderabad | 225000.00 |
| Vijayawada | 185000.00 |
| Bangalore | 120000.00 |
| Mumbai | 150000.00 |
+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> CREATE VIEW Deposit_Transaction_View AS
-> SELECT *
-> FROM Bank_Transaction
-> WHERE Transaction_Type = 'DEPOSIT';
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> SELECT * FROM Deposit_Transaction_View;
+-----+-----+-----+-----+-----+
| Transaction_ID | Account_No | Transaction_Type | Amount | Transaction_Date |
+-----+-----+-----+-----+-----+
| 1 | 10001 | DEPOSIT | 10000.00 | 2026-08-15 16:02:47 |
| 3 | 10002 | DEPOSIT | 15000.00 | 2026-08-15 16:02:47 |
| 5 | 10004 | DEPOSIT | 5000.00 | 2026-08-15 16:02:47 |
| 7 | 10006 | DEPOSIT | 12000.00 | 2026-08-15 16:02:47 |
| 8 | 10007 | DEPOSIT | 25000.00 | 2026-08-15 16:02:47 |
| 10 | 10009 | DEPOSIT | 20000.00 | 2026-08-15 16:02:47 |
+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

```
mysql> CREATE VIEW Withdrawal_Transaction_View AS
-> SELECT *
-> FROM Bank_Transaction
-> WHERE Transaction_Type = 'WITHDRAW';
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> SELECT * FROM Withdrawal_Transaction_View;
```

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
2	10001	WITHDRAW	5000.00	2026-08-15 16:02:47
4	10003	WITHDRAW	20000.00	2026-08-15 16:02:47
6	10005	WITHDRAW	10000.00	2026-08-15 16:02:47
9	10008	WITHDRAW	5000.00	2026-08-15 16:02:47
11	10010	WITHDRAW	15000.00	2026-08-15 16:02:47

5 rows in set (0.00 sec)

```
mysql> CREATE VIEW Customer_Transaction_View AS
```

```
--> SELECT
-->     C.Customer_Name,
-->     A.Account_No,
-->     A.Account_Type,
-->     T.Transaction_ID,
-->     T.Transaction_Type,
-->     T.Amount,
-->     T.Transaction_Date
--> FROM Customer C
--> JOIN Account A
--> ON C.Customer_ID = A.Customer_ID
--> JOIN Bank_Transaction T
--> ON A.Account_No = T.Account_No;
```

Query OK, 0 rows affected (0.02 sec)

```
mysql> SELECT * FROM Customer_Transaction_View;
```

Customer_Name	Account_No	Account_Type	Transaction_ID	Transaction_Type	Amount	Transaction_Date
Ravi Kumar	10001	Savings	1	DEPOSIT	10000.00	2026-08-15 16:02:47
Ravi Kumar	10001	Savings	2	WITHDRAW	5000.00	2026-08-15 16:02:47
Priya Sharma	10002	Savings	3	DEPOSIT	15000.00	2026-08-15 16:02:47
Arjun Reddy	10003	Current	4	WITHDRAW	20000.00	2026-08-15 16:02:47
Sneha Rao	10004	Savings	5	DEPOSIT	5000.00	2026-08-15 16:02:47
Kiran Kumar	10005	Current	6	WITHDRAW	10000.00	2026-08-15 16:02:47
Anil Kumar	10006	Savings	7	DEPOSIT	12000.00	2026-08-15 16:02:47
Meena Reddy	10007	Current	8	DEPOSIT	25000.00	2026-08-15 16:02:47
Rahul Sharma	10008	Savings	9	WITHDRAW	5000.00	2026-08-15 16:02:47
Lakshmi Devi	10009	Savings	10	DEPOSIT	20000.00	2026-08-15 16:02:47
Suresh Babu	10010	Current	11	WITHDRAW	15000.00	2026-08-15 16:02:47

11 rows in set (0.00 sec)

```
mysql> CREATE VIEW High_Value_Transaction_View AS
```

```
--> SELECT *
--> FROM Bank_Transaction
--> WHERE Amount > 10000;
```

Query OK, 0 rows affected (0.02 sec)

```
mysql> SELECT * FROM High_Value_Transaction_View;
```

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
3	10002	DEPOSIT	15000.00	2026-08-15 16:02:47
4	10003	WITHDRAW	20000.00	2026-08-15 16:02:47
7	10006	DEPOSIT	12000.00	2026-08-15 16:02:47
8	10007	DEPOSIT	25000.00	2026-08-15 16:02:47
10	10009	DEPOSIT	20000.00	2026-08-15 16:02:47
11	10010	WITHDRAW	15000.00	2026-08-15 16:02:47

```
mysql> CREATE VIEW Balance_Ranking_View AS
-> SELECT
->     Account_No,
->     Customer_ID,
->     Account_Type,
->     Balance
-> FROM Account
-> ORDER BY Balance DESC;
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> SELECT * FROM Balance_Ranking_View;
+-----+-----+-----+-----+
| Account_No | Customer_ID | Account_Type | Balance |
+-----+-----+-----+-----+
| 10007 | 107 | Current | 150000.00 |
| 10003 | 103 | Current | 120000.00 |
| 10010 | 110 | Current | 110000.00 |
| 10005 | 105 | Current | 90000.00 |
| 10009 | 109 | Savings | 85000.00 |
| 10002 | 102 | Savings | 75000.00 |
| 10006 | 106 | Savings | 65000.00 |
| 10001 | 101 | Savings | 50000.00 |
| 10004 | 104 | Savings | 45000.00 |
| 10008 | 108 | Savings | 35000.00 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

```
mysql> CREATE VIEW Customer_Name_View AS
-> SELECT
->     Customer_ID,
->     Customer_Name,
->     City
-> FROM Customer
-> ORDER BY Customer_Name;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT * FROM Customer_Name_View;
+-----+-----+-----+
| Customer_ID | Customer_Name | City |
+-----+-----+-----+
| 106 | Anil Kumar | Delhi |
| 103 | Arjun Reddy | Bangalore |
| 105 | Kiran Kumar | Hyderabad |
| 109 | Lakshmi Devi | Hyderabad |
| 107 | Meena Reddy | Mumbai |
| 102 | Priya Sharma | Vijayawada |
| 108 | Rahul Sharma | Pune |
| 101 | Ravi Kumar | Hyderabad |
| 104 | Sneha Rao | Chennai |
| 110 | Suresh Babu | Vijayawada |
+-----+-----+-----+
10 rows in set (0.02 sec)
```

```
mysql> CREATE VIEW Loan_Interest_View AS
-> SELECT
->     Loan_ID,
->     Customer_ID,
->     Loan_Type,
->     Loan_Amount,
->     Interest_Rate,
->     (Loan_Amount * Interest_Rate / 100) AS Annual_Interest
-> FROM Loan;
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> SELECT * FROM Loan_Interest_View;
+-----+-----+-----+-----+-----+-----+
| Loan_ID | Customer_ID | Loan_Type | Loan_Amount | Interest_Rate | Annual_Interest |
+-----+-----+-----+-----+-----+-----+
| 501 | 101 | Home Loan | 5000000.00 | 7.50 | 375000.00000000 |
| 502 | 102 | Education Loan | 1000000.00 | 6.50 | 65000.00000000 |
| 503 | 103 | Car Loan | 800000.00 | 8.20 | 65600.00000000 |
+-----+-----+-----+-----+-----+-----+
```

```
mysql> SELECT * FROM Loan_Interest_View;
```

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate	Annual_Interest
501	101	Home Loan	5000000.00	7.50	375000.00000000
502	102	Education Loan	1000000.00	6.50	65000.00000000
503	103	Car Loan	800000.00	8.20	65600.00000000
504	104	Personal Loan	500000.00	10.50	52500.00000000
505	105	Home Loan	4000000.00	7.20	288000.00000000
506	106	Car Loan	900000.00	8.50	76500.00000000
507	107	Business Loan	3000000.00	9.00	270000.00000000
508	109	Personal Loan	600000.00	10.00	60000.00000000

8 rows in set (0.00 sec)

```
mysql> CREATE VIEW Loan_Total_Amount_View AS
```

```
--> SELECT
-->   Loan_ID,
-->   Customer_ID,
-->   Loan_Type,
-->   Loan_Amount,
-->   Interest_Rate,
-->   Loan_Amount +
-->   (Loan_Amount * Interest_Rate / 100)
-->   AS Total_Amount
--> FROM Loan;
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> SELECT * FROM Loan_Total_Amount_View;
```

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate	Total_Amount
501	101	Home Loan	5000000.00	7.50	5375000.00000000
502	102	Education Loan	1000000.00	6.50	1065000.00000000
503	103	Car Loan	800000.00	8.20	865600.00000000
504	104	Personal Loan	500000.00	10.50	552500.00000000

```
mysql> SELECT * FROM Loan_Total_Amount_View;
```

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate	Total_Amount
501	101	Home Loan	5000000.00	7.50	5375000.00000000
502	102	Education Loan	1000000.00	6.50	1065000.00000000
503	103	Car Loan	800000.00	8.20	865600.00000000
504	104	Personal Loan	500000.00	10.50	552500.00000000
505	105	Home Loan	4000000.00	7.20	4288000.00000000
506	106	Car Loan	900000.00	8.50	976500.00000000
507	107	Business Loan	3000000.00	9.00	3270000.00000000
508	109	Personal Loan	600000.00	10.00	660000.00000000

8 rows in set (0.00 sec)

```
mysql> SELECT *
--> FROM High_Balance_View
--> WHERE Balance > 120000;
```

Account_No	Customer_ID	Account_Type	Balance
10007	107	Current	150000.00

1 row in set (0.00 sec)

```
mysql> SELECT *
--> FROM Savings_Account_View
--> WHERE Balance > 60000;
```

Account_No	Customer_ID	Account_Type	Balance	Branch
10002	102	Savings	75000.00	Vijayawada
10006	106	Savings	65000.00	Delhi
10009	109	Savings	85000.00	Hyderabad

```
mysql> SELECT *
-> FROM Hyderabad_Account_View
-> WHERE Balance > 50000;
+-----+-----+-----+-----+-----+
| Account_No | Customer_ID | Account_Type | Balance | Branch |
+-----+-----+-----+-----+-----+
| 10005 | 105 | Current | 90000.00 | Hyderabad |
| 10009 | 109 | Savings | 85000.00 | Hyderabad |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> SELECT *
-> FROM Customer_Basic_View
-> WHERE City = 'Hyderabad';
+-----+-----+-----+
| Customer_ID | Customer_Name | City |
+-----+-----+-----+
| 101 | Ravi Kumar | Hyderabad |
| 105 | Kiran Kumar | Hyderabad |
| 109 | Lakshmi Devi | Hyderabad |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT *
-> FROM Customer_Account_View
-> ORDER BY Balance DESC;
+-----+-----+-----+-----+-----+-----+-----+
| Customer_ID | Customer_Name | City | Account_No | Account_Type | Balance | Branch |
+-----+-----+-----+-----+-----+-----+-----+
| 107 | Meena Reddy | Mumbai | 10007 | Current | 150000.00 | Mumbai |
| 103 | Arjun Reddy | Bangalore | 10003 | Current | 120000.00 | Bangalore |
| 110 | Suresh Babu | Vijayawada | 10010 | Current | 110000.00 | Vijayawada |
| 105 | Kiran Kumar | Hyderabad | 10005 | Current | 90000.00 | Hyderabad |
| 109 | Lakshmi Devi | Hyderabad | 10009 | Savings | 85000.00 | Hyderabad |
| 102 | Priya Sharma | Vijayawada | 10002 | Savings | 75000.00 | Vijayawada |
| 106 | Anil Kumar | Delhi | 10006 | Savings | 65000.00 | Delhi |
| 102 | Priya Sharma | Vijayawada | 10002 | Savings | 75000.00 | Vijayawada |
| 106 | Anil Kumar | Delhi | 10006 | Savings | 65000.00 | Delhi |
| 101 | Ravi Kumar | Hyderabad | 10001 | Savings | 50000.00 | Hyderabad |
| 104 | Sneha Rao | Chennai | 10004 | Savings | 45000.00 | Chennai |
| 108 | Rahul Sharma | Pune | 10008 | Savings | 35000.00 | Pune |
+-----+-----+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql> SELECT *
-> FROM Customer_Loan_View
-> WHERE Loan_Amount > 1000000;
+-----+-----+-----+-----+-----+-----+-----+
| Customer_ID | Customer_Name | City | Loan_ID | Loan_Type | Loan_Amount | Interest_Rate |
+-----+-----+-----+-----+-----+-----+-----+
| 101 | Ravi Kumar | Hyderabad | 501 | Home Loan | 5000000.00 | 7.50 |
| 105 | Kiran Kumar | Hyderabad | 505 | Home Loan | 4000000.00 | 7.20 |
| 107 | Meena Reddy | Mumbai | 507 | Business Loan | 3000000.00 | 9.00 |
+-----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> UPDATE Account_View
-> SET Balance = 60000
-> WHERE Account_No = 10001;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT *
-> FROM Account
-> WHERE Account_No = 10001;
+-----+-----+-----+-----+-----+
| Account_No | Customer_ID | Account_Type | Balance | Branch |
+-----+-----+-----+-----+-----+
| 10001 | 101 | Savings | 60000.00 | Hyderabad |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> CREATE VIEW Simple_Account_View AS
```



```
mysql> CREATE VIEW Simple_Account_View AS
-> SELECT
->     Account_No,
->     Customer_ID,
->     Account_Type,
->     Balance,
->     Branch
-> FROM Account;
Query OK, 0 rows affected (0.02 sec)

mysql> INSERT INTO Simple_Account_View
-> VALUES
-> (10011, 101, 'Savings', 55000, 'Hyderabad');
Query OK, 1 row affected (0.02 sec)
```

```
mysql> SELECT * FROM Account;
+-----+-----+-----+-----+-----+
| Account_No | Customer_ID | Account_Type | Balance | Branch |
+-----+-----+-----+-----+-----+
| 10001 | 101 | Savings | 60000.00 | Hyderabad |
| 10002 | 102 | Savings | 75000.00 | Vijayawada |
| 10003 | 103 | Current | 120000.00 | Bangalore |
| 10004 | 104 | Savings | 45000.00 | Chennai |
| 10005 | 105 | Current | 90000.00 | Hyderabad |
| 10006 | 106 | Savings | 65000.00 | Delhi |
| 10007 | 107 | Current | 150000.00 | Mumbai |
| 10008 | 108 | Savings | 35000.00 | Pune |
| 10009 | 109 | Savings | 85000.00 | Hyderabad |
| 10010 | 110 | Current | 110000.00 | Vijayawada |
| 10011 | 101 | Savings | 55000.00 | Hyderabad |
+-----+-----+-----+-----+-----+
11 rows in set (0.00 sec)
```

```
mysql> DELETE FROM Simple_Account_View
-> WHERE Account_No = 10011;
Query OK, 1 row affected (0.02 sec)
```

```
mysql> SELECT * FROM Account;
+-----+-----+-----+-----+-----+
| Account_No | Customer_ID | Account_Type | Balance | Branch |
+-----+-----+-----+-----+-----+
| 10001 | 101 | Savings | 60000.00 | Hyderabad |
| 10002 | 102 | Savings | 75000.00 | Vijayawada |
| 10003 | 103 | Current | 120000.00 | Bangalore |
| 10004 | 104 | Savings | 45000.00 | Chennai |
| 10005 | 105 | Current | 90000.00 | Hyderabad |
| 10006 | 106 | Savings | 65000.00 | Delhi |
| 10007 | 107 | Current | 150000.00 | Mumbai |
| 10008 | 108 | Savings | 35000.00 | Pune |
| 10009 | 109 | Savings | 85000.00 | Hyderabad |
| 10010 | 110 | Current | 110000.00 | Vijayawada |
+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

```
mysql> SHOW FULL TABLES
-> WHERE TABLE_TYPE = 'VIEW';
```

```
+-----+-----+
| Tables_in_bankviewsdb | Table_type |
+-----+-----+
| account_count_view    | VIEW      |
| account_type_balance  | VIEW      |
| account_type_count    | VIEW      |
| account_view          | VIEW      |
| average_account_balance | VIEW      |
| balance_ranking_view  | VIEW      |
| branch_account_count  | VIEW      |
| branch_total_balance  | VIEW      |
| current_account_view  | VIEW      |
| customer_account_view | VIEW      |
| customer_banking_view | VIEW      |
| customer_basic_view   | VIEW      |
| customer_loan_view    | VIEW      |
+-----+-----+
```

```

+-----+-----+
| Tables_in_bankviewsdb | Table_type |
+-----+-----+
| account_count_view    | VIEW      |
| account_type_balance  | VIEW      |
| account_type_count    | VIEW      |
| account_view          | VIEW      |
| average_account_balance | VIEW      |
| balance_ranking_view  | VIEW      |
| branch_account_count  | VIEW      |
| branch_total_balance  | VIEW      |
| current_account_view  | VIEW      |
| customer_account_view | VIEW      |
| customer_banking_view | VIEW      |
| customer_basic_view   | VIEW      |
| customer_loan_view    | VIEW      |
| customer_name_view    | VIEW      |
| customer_transaction_view | VIEW      |
| customer_view         | VIEW      |
| deposit_transaction_view | VIEW      |
| high_balance_view     | VIEW      |
| high_value_transaction_view | VIEW      |
| hyderabad_account_view | VIEW      |
| hyderabad_customer_accounts | VIEW      |
| loan_interest_view    | VIEW      |
| loan_total_amount_view | VIEW      |
| low_balance_view      | VIEW      |
| maximum_balance_view  | VIEW      |
| minimum_balance_view  | VIEW      |
| multiple_account_branches | VIEW      |
| rich_branches         | VIEW      |
| savings_account_view  | VIEW      |
| simple_account_view   | VIEW      |
| total_bank_balance    | VIEW      |
| withdrawal_transaction_view | VIEW      |
+-----+-----+
32 rows in set (0.06 sec)

```

```
mysql> SHOW CREATE VIEW Customer_Account_View;
```

```

+-----+-----+
| View | Create View |
+-----+-----+
| character_set_client | collation_connection |
+-----+-----+
| customer_account_view | CREATE ALGORITHM=UNDEFINED DEFINER='root'@'localhost' SQL SECURITY DEFINER VIEW 'customer_account_view' AS
select 'c'.Customer_ID AS 'Customer_ID','c'.Customer_Name AS 'Customer_Name','c'.City AS 'City','a'.Account_No AS 'Account_No',
'a'.Account_Type AS 'Account_Type','a'.Balance AS 'Balance','a'.Branch AS 'Branch' from ('customer' 'c' join 'account' 'a' on
('c'.Customer_ID = 'a'.Customer_ID))) | cp850 | cp850_general_ci |
+-----+-----+
1 row in set (0.00 sec)

```

```
mysql> DESCRIBE Customer_Account_View;
```

```

+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Customer_ID | int | NO | | NULL | |
| Customer_Name | varchar(100) | NO | | NULL | |
| City | varchar(50) | YES | | NULL | |
| Account_No | int | NO | | NULL | |
| Account_Type | varchar(20) | YES | | NULL | |
| Balance | decimal(12,2) | YES | | NULL | |
| Branch | varchar(50) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.03 sec)

```

```
mysql> DROP VIEW Customer_View;
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> DROP VIEW
-> Customer_Basic_View,
-> Savings_Account_View,
-> Current_Account_View;
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> |
```